

2. FinOps and Cloud Economics

2.1 - FinOps as a Business and Operational Framework

FinOps provides the business framework for interpreting the project's cost data. A pipeline can reconcile charges technically while leaving allocation, ownership or business meaning unresolved. The platform therefore supplies data for financial questions as well as for reporting.

The FinOps Foundation framework defines FinOps through collaboration between engineering, finance and business functions, timely data and financial accountability. The project applies this framework to a bounded problem: preparing governed cost products and examining what decisions those products can support. It does not evaluate the maturity of the organization's full FinOps practice.

Source: FinOps Foundation, FinOps Framework.

Why cloud economics require FinOps

Cloud expenditure varies with usage, configuration, location and contractual terms. Resource operators, budget owners and invoice reviewers may therefore need different views of the same charge. Shared cost definitions and explicit allocation rules allow these views to be compared without treating every difference as waste or saving.

Roles in the proposed operating model

- The Data Platform function provides ingestion, transformation, quality, security and operational controls.
- FinOps owns cost semantics, allocation rules, financial indicators and business validation.
- The Power BI or dashboard function owns presentation and semantic-model standards.
- Managers and application owners consume information for their authorized perimeter.
- Authorized analysts may need direct access to governed datamarts for additional analysis.

This is a design allocation of responsibility. The existing dashboard's actual permission and change processes are described from the student's account in section 1.2; the proposed roles require organizational validation before they can be reported as established practice.

2.2 - Inform, Optimize and Operate in the Project

Analytical positioning

The FinOps Foundation describes Inform, Optimize and Operate as recurring phases. This thesis uses them to organize the questions tested by the prototype, while recognizing that a reliable cost measure can inform several activities at once.

Source: FinOps Foundation, FinOps Phases.

Inform

Inform establishes visibility and accountability. It includes cost trends, allocation, showback, comparison between periods, identification of unallocated expenditure and access to consistent definitions. In the prototype, Inform is represented principally by the certified datamarts and the Streamlit dashboard.

Optimize

Optimize evaluates whether expenditure delivers sufficient value and whether rates, commitments or resource usage can be improved. The current dataset supports comparisons among list, contracted, effective and billed cost. It does not by itself support defensible rightsizing recommendations because those require utilization, inventory, recommendation or pricing-reference data.

Operate

Operate embeds FinOps into recurring processes. For this PFE it includes ingestion, data quality, reconciliation, freshness, execution traceability and recovery. Forecasting and anomaly detection remain conditional on sufficient historical data and an explicit evaluation protocol.

Boundary of claims

The dashboard demonstrates cost visibility on synthetic data. It does not demonstrate measured organizational savings, a reduction in company workload or production-grade continuous operation. These claims require separate evidence.

2.3 - Cloud Pricing and Cost Mechanisms

From consumption to a financial charge

A cloud charge results from the interaction of a consumed service, a pricing quantity and unit, a public rate, contractual conditions, commitment benefits, credits, corrections, taxes and billing timing. Consequently, the word “cost” is ambiguous unless the selected cost perspective and accounting basis are stated.

At line level, the simplified pricing path used for interpretation is:

```

Consumed resource or service
    v conversion to pricing quantity
Public list price
    v negotiated contractual reductions
Contracted price
    v allocation of commitment benefits and amortization
Effective cost
    v invoice timing and covering-charge treatment
Billed cost
  
```

This diagram is explanatory. The equality and ordering of these values cannot be assumed for every charge category, credit, correction, tax or purchase row.

Four FOCUS cost perspectives

Measure	Business interpretation	Main use in the project
ListCost	Cost calculated from the provider's public list unit price and pricing quantity	Reference for rate-comparison analysis
ContractedCost	Cost calculated from the negotiated contracted unit price and pricing quantity	Isolate the effect of negotiated pricing where comparable
EffectiveCost	Economically allocated cost after relevant discounts and amortized commitment purchases	Accrual-oriented service and resource analysis
BilledCost	Charge serving as the basis for invoicing in the billing period	Invoice reconciliation, cash-oriented reporting and allocation

These definitions follow the FOCUS 1.0 specifications for List Cost, Contracted Cost, Effective Cost and Billed Cost. BilledCost and EffectiveCost answer different questions. A commitment purchase may be billed at one time while its economic cost is allocated to covered usage over time. Their difference must therefore not automatically be labelled as a saving.

Source: FOCUS Project, FOCUS 1.0: List Cost, Contracted Cost, Effective Cost and Billed Cost.

Pricing quantity and unit price

Where the required fields are available and comparable:

```

ListCost      = ListUnitPrice      × PricingQuantity
ContractedCost = ContractedUnitPrice × PricingQuantity
  
```

Pricing quantity can differ from consumed quantity because providers may apply block, tier or normalized pricing. Corrections are also capable of breaking a naive equality at individual-row level and must be treated according to their charge class.

Negotiated and commitment-related effects

The following decompositions are analytical candidates, not universally valid accounting identities:

```
Negotiated rate effect      = ListCost - ContractedCost
Additional effective effect = ContractedCost - EffectiveCost
Total effective rate effect = ListCost - EffectiveCost
```

They may be calculated only on a consistent currency, period, scope and charge population. Purchase and covered-usage records must be filtered according to a documented billed or accrual basis to avoid double counting.

Commitment discounts

Reservations and Savings Plans exchange an obligation to consume or spend over a defined period for reduced rates. Commitment purchases may be upfront, recurring or mixed. Their cost is amortized across charge periods and allocated to eligible usage. Unused commitment is a cost of underutilization, not an additional saving, and follows a use-it-or-lose-it logic.

The prototype contains commitment-related dimensions and cost measures, but it must not claim complete commitment utilization analysis until identifiers, statuses, purchase rows, covered usage and unused portions have been validated in the dataset.

Realized versus potential savings

- A realized rate effect can be derived retrospectively from comparable populated cost measures, with explicit exclusions.
- A potential saving estimates a future action and normally requires additional evidence such as utilization metrics, resource inventory, provider recommendations, price references or commitment coverage.

The current dashboard's `ListCost - EffectiveCost` indicator should therefore be described as an effective difference from list cost until its scope and interpretation are fully validated. It must not be presented as a verified business saving without that validation.

2.4 - Allocation, Showback and Governed Self-Service

Allocation and accountability

Allocation associates technology expenditure with an accountable business perimeter such as a domain, application, project, team or cost center. Unallocated costs must remain visible rather than being silently removed because they are themselves a governance signal.

Source: FinOps Foundation, Allocation.

Showback and chargeback

Showback exposes the cost attributed to a consumer without necessarily performing an internal financial transfer. Chargeback goes further by assigning or recovering that cost through an internal accounting mechanism. The current PFE implements a showback-oriented decision-support scenario; it does not implement an enterprise chargeback process.

Source: FinOps Foundation, Invoicing & Chargeback.

Governed self-service

The proposed access model gives managers information for their authorized perimeter, FinOps consolidated visibility and definition ownership, and authorized analysts read access to certified datamarts. Dashboard-level filtering requires corresponding controls on direct data access. The project has not yet validated those controls with separate user identities.

Connection to the professional problem

The project framing identifies repeated requests and manual extracts as possible sources of effort. Their frequency and cost have not been measured. A self-service scenario can test whether users complete specified analyses without a new FinOps extract, but an organizational benefit would require evidence from the actual workflow.

References

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